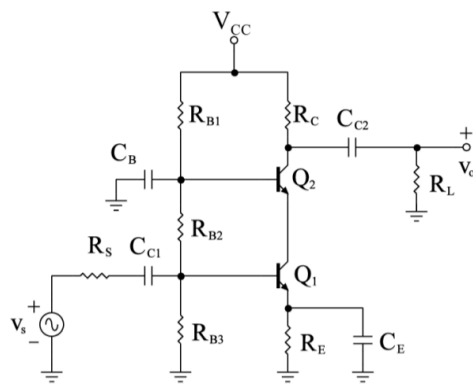


ELEC 301 Project 2

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Problem 1: Multi-transistors amplifier circuits

A cascode amplifier is a two-stage amplifier that combines a common-emitter stage with a common-base stage, to achieve high bandwidth and voltage gain while minimizing the Miller effect. The following cascode circuit implementation will be used for your design:



Introduce the circuit schematics into your Spice simulator software, using the 2N3904 model for the transistors. The minimum specifications for this amplifier are presented below:

R_{out} (value at midband)	R_{in} (minimum value at midband) ¹	$ A_M $ (minimum value at midband)	$\omega_{L,3dB}$ (maximum value for low-frequency cut-in)
$2.5k\Omega \pm 250\Omega$	$3.5k\Omega$	50V/V	1200 rad/s

Assume the source resistance $R_s=50\Omega$, that $V_{cc}=20V$ and that the output load resistance $R_L=50k\Omega$. You are required to design the cascode circuit to meet the above specifications.

Start by calculating the resistor values needed to bias the transistors Q1 and Q2. Once you have computed the values of the bias resistors, change them to closest commonly available resistance values. Assuming a large capacitance value for C_B (use $C_B=200mF$), you should obtain values for all of the other capacitances (use commonly/standard available values in your final design). Compute the small-signal parameters for your circuit.

- Measure the DC operating point for the cascode amplifier
- Plot the Bode plots for magnitude and phase and compare your estimates of the locations of ω_{L3dB} and ω_{H3dB} with your calculated values.

- C. Using the magnitude Bode plot, pick a midband frequency. Using this frequency for the input voltage, adjust the amplitude of the input signal until you see that the output signal (viewed in the time domain) is becoming non-linear (you can vary the amplitude of the input signal, at the constant frequency, and plot the voltage transfer curve, the amplitude of v_o vs. v_s).
- D. Measure the input impedance of your amplifier at midband (not including the source resistor $R_s=50\Omega$) and compare this with the input impedance that you designed for.

Please include your comparison between measurements and your computations, your interpretations and insights.

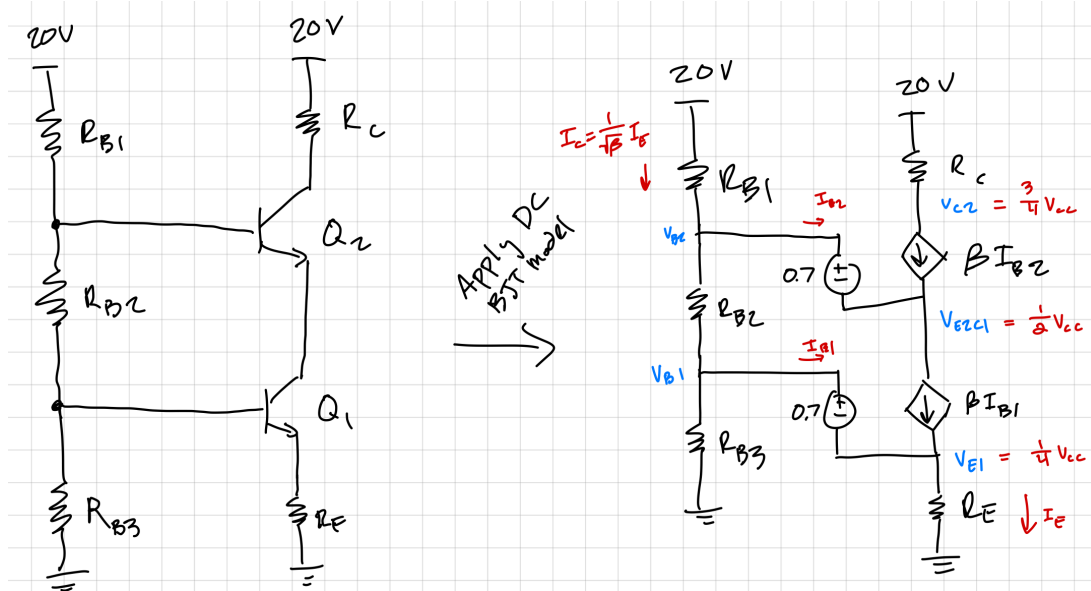
Designing the cascode circuit

For the 2N3904 transistor model in LTSpice we have the following spice directive:

```
.model 2N3904 NPN(IS=1E-14 VAF=100 Bf=300 IKF=0.4 XTB=1.5 BR=4 CJC=4E-12
CJE=8E-12 RB=20 RC=0.1 RE=0.1 TR=250E-9 TF=350E-12 ITF=1 VTF=2 XTF=3
Vceo=40 Icrating=200m mfg=NXP)
```

We see here that $\beta = 300$ and $V_A = 100\text{ V}$, which will be helpful when calculating our values

To get the values of the resistors we need to observe the DC model. All capacitors effectively become open circuits in the steady state and we can apply our DC model of the BJT. We can also apply a variation of the $\frac{1}{3}$ rule (the $\frac{1}{4}$ rule) for the cascode amplifier to evenly split V_{CC} across the BJTs as shown below to simplify our calculations and approximate the needed voltages to bias the transistors. We can also approximate that $I_C = I_E \frac{1}{\sqrt{\beta}}$ to further simplify our circuit and save us time:



Utilizing these simplifications we develop the following list of equations:

$$I_E \frac{1}{\sqrt{\beta}} = \left(\frac{\frac{20}{4}}{R_E}\right) \frac{1}{\sqrt{\beta}} = \frac{20 - (\frac{20}{2} + 0.7)}{R_{B1}} \rightarrow \frac{5}{R_E \sqrt{300}} = \frac{9.3}{R_{B1}}$$

$$\left(\frac{\frac{20}{4}}{R_E}\right) \frac{1}{\sqrt{\beta}} = I_{B2} + \frac{(\frac{20}{2} + 0.7) - (\frac{20}{4} + 0.7)}{R_{B2}} \rightarrow \frac{5}{R_E \sqrt{300}} = I_{B2} + \frac{5}{R_{B2}}$$

$$\frac{(\frac{20}{2} + 0.7) - (\frac{20}{4} + 0.7)}{R_{B2}} = I_{B1} + \frac{\frac{20}{4} + 0.7}{R_{B3}} \rightarrow \frac{5}{R_{B2}} = I_{B1} + \frac{5.7}{R_{B3}}$$

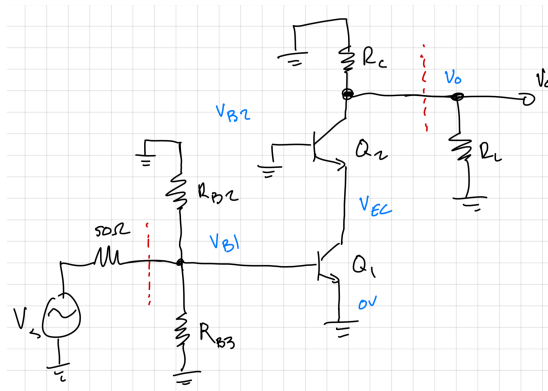
$$\frac{20 - \frac{3 \cdot 20}{4}}{R_C} = \beta I_{B2} \rightarrow \frac{5}{R_C} = 300 I_{B2}$$

$$(1 + \beta) I_{B2} = \beta I_{B1} \rightarrow 301 I_{B2} = 300 I_{B1}$$

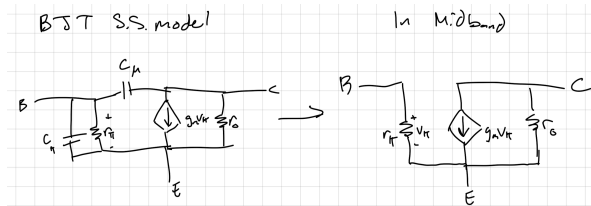
$$(1 + \beta) I_{B1} = \frac{\frac{20}{4}}{R_E} \rightarrow 301 I_{B1} = \frac{5}{R_E}$$

Our unknowns here are I_{B1} , I_{B2} , R_{B1} , R_{B2} , R_{B3} , R_C , and R_E , for a total of 7 unknowns in a list of 6 equations.

We can get the last expression from the midband circuit, where the low frequency capacitors short and the high frequency capacitors open giving us:

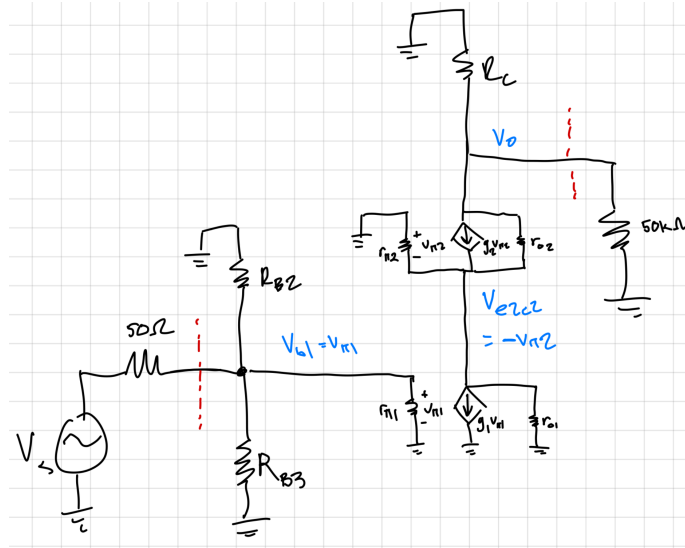


We can use our small signal model of the BJT:



Where $r_\pi = \frac{V_T}{I_B} = \beta \frac{V_T}{I_C}$, $r_o = \frac{V_A}{I_C}$, and $g_m = \frac{I_C}{V_T}$. We assume $V_T = 25\text{mV}$ @ 25°C .

This gives us the following circuit:



We can now refer to our design specifications to observe that, for the output resistance value, the $\pm 250\Omega$ gives us some leeway with determining the correct values. We know that both of the r_o values are quite large, and as such the resulting output resistance in the midband will mostly depend on the value of R_C as it'll be significantly smaller than the resistance of the rest of the midband circuit that is connected in parallel, meaning it'll "dominate" the final value. This means we can simply set R_C to be in the middle of our midband output resistance value, or:

$$R_C = 2.5 \text{ k}\Omega$$

With that as our final expression, we can solve the DC circuit to give us the resistor values:

$$R_{B1} = 80.006 \text{ k}\Omega$$

$$R_{B2} = 45.631 \text{ k}\Omega$$

$$R_{B3} = 55.401 \text{ k}\Omega$$

$$R_E = 2.4834 \Omega$$

And the DC current values which we will need to calculate some circuit parameters for the midband:

$$I_{B1} = \frac{301}{15000000} = 6.6889 \mu\text{A}$$

$$I_{B2} = 6.6667 \mu\text{A}$$

$$I_{C1} = \beta I_{B1} = 2.0067 \text{ mA}$$

$$I_{C2} = \beta I_{B2} = 2.0000 \text{ mA}$$

As per the specifications, we need to change the resistor values to the closest commonly available resistance values. Referring to

<https://www.open.edu/openlearn/science-maths-technology/an-introduction-electronics/content-section-2.4>, we see that the closest matching common resistor values to each of the resistors we found are as follows:

$$R_{B1} = 82 \text{ k}\Omega$$

$$R_{B2} = 47 \text{ k}\Omega$$

$$R_{B3} = 56 \text{ k}\Omega$$

$$R_E = 2.4 \text{ k}\Omega$$

$$R_C = 2.4 \text{ k}\Omega$$

With the DC current values we can calculate the small signal parameters as shown below. There's no point in completely recomputing the currents for the new values since they should result in similar values and we're already making many assumptions.

$$r_{\pi1} = \frac{V_T}{I_{B1}} = 3.7375 \text{ k}\Omega$$

$$r_{\pi2} = \frac{V_T}{I_{B2}} = 3.7500 \text{ k}\Omega$$

$$r_{o1} = \frac{V_A}{I_{C1}} = \frac{V_A}{\beta I_{B1}} = 49.834 \text{ k}\Omega$$

$$r_{o2} = \frac{V_A}{\beta I_{B2}} = 50.000 \text{ k}\Omega$$

$$g_1 = \frac{I_{C1}}{V_T} = \frac{\beta I_{B1}}{V_T} = 0.080267$$

$$g_2 = \frac{\beta I_{B2}}{V_T} = 0.08$$

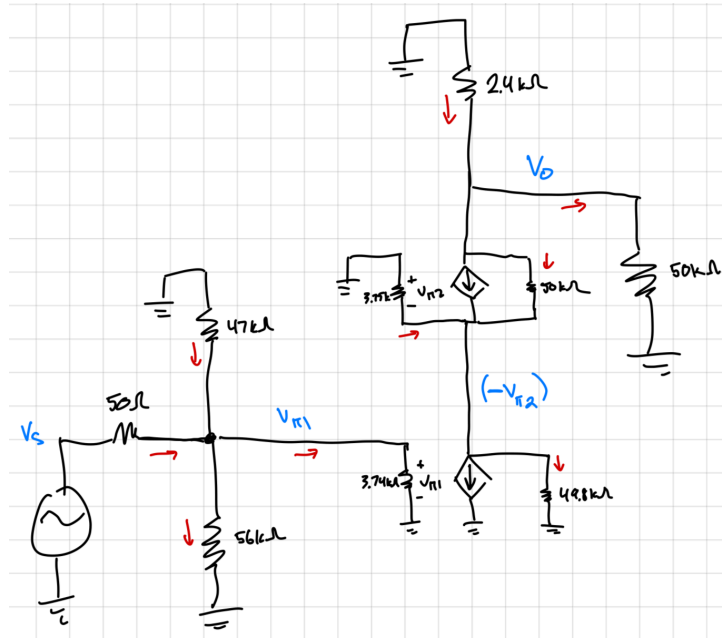
Our small signal capacitor parameter values can be found from the LTspice model parameters:

$$C_{\mu1} = C_{\mu2} = CJC = 4 \text{ pF}$$

$$C_{\pi1} = CJE + g_1 TF = 8 \text{ pF} + 0.080267 \cdot 10^{-12} = 8.0803 \text{ pF}$$

$$C_{\pi2} = CJE + g_2 TF = 8 \text{ pF} + 0.08 \cdot 10^{-12} = 8.08 \text{ pF}$$

Now we can revisit our design specifications and correct any errors:



Recall the specifications:

R_{out} (value at midband)	R_{in} (minimum value at midband) ¹	$ A_M $ (minimum value at midband)	$\omega_{L,3dB}$ (maximum value for low-frequency cut-in)
$2.5k\Omega \pm 250\Omega$	$3.5k\Omega$	$50V/V$	1200 rad/s

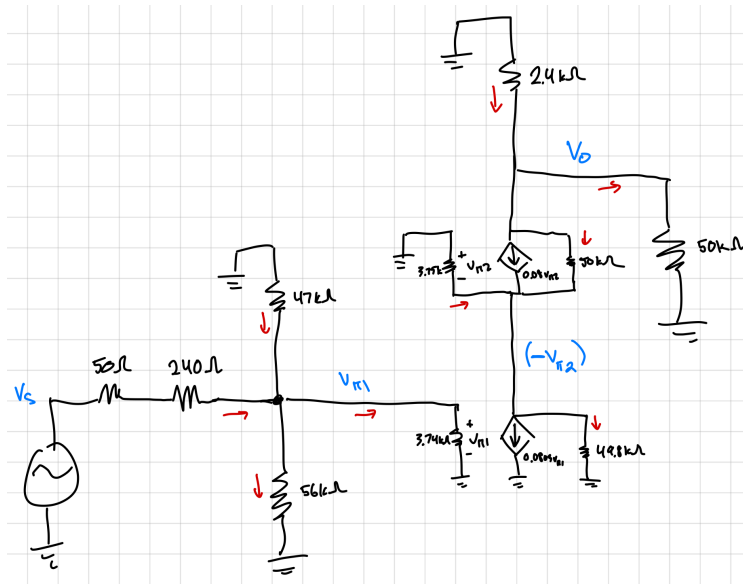
We see that for the input resistance in the midband circuit:

$$R_{in}(midband) = R_{B2} // R_{B3} // r_{\pi 1} = 3261\Omega \neq 3500\Omega$$

To fix this, we need to add a resistor at the input following the C_{C1} capacitor, and solve:

$$R_{in}(midband) = R_{input} + (R_{B2} // R_{B3} // r_{\pi 1}) = 3500\Omega \rightarrow R_{input} = 239.7\Omega$$

We need to simplify this value to the nearest common resistance value that doesn't put us below the $3.5k\Omega$ requirement, which can be $R_{input} = 240\Omega$



Next, we require that $|A_M| = \left| \frac{v_o}{v_s} \right| \geq 50$. To verify we can quickly solve the circuit now that we have all the parameters:

$$\frac{v_s - v_{\pi1}}{50 + 240} + \frac{0 - v_{\pi1}}{47000} = \frac{v_{\pi1}}{56000} + \frac{v_{\pi1}}{3738}$$

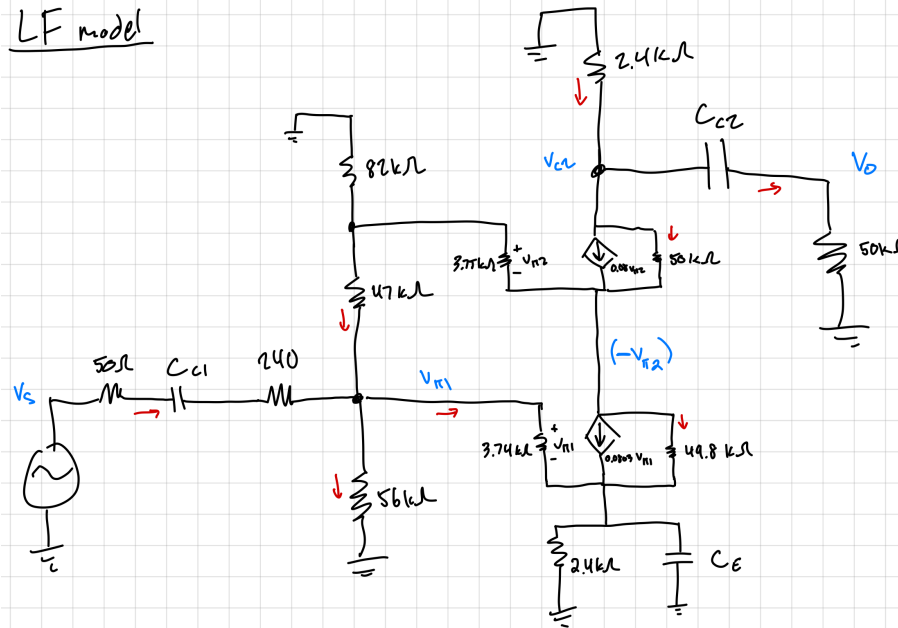
$$\frac{0 - v_o}{2400} = \frac{v_o}{50000} + 0.08v_{\pi2} + \frac{v_o - (-v_{\pi2})}{50000}$$

$$\frac{0 - (-v_{\pi2})}{3750} + 0.08v_{\pi2} + \frac{v_o - (-v_{\pi2})}{50000} = 0.0803v_{\pi1} + \frac{-v_{\pi2}}{49800}$$

Solving for the expression of $\frac{v_o}{v_s}$ gives us $\left| \frac{v_o}{v_s} \right| = \sim 168$ which is definitely greater than 50, meaning our parameters work.

Lastly, we need to determine the capacitance values given the final specification, $\omega_{L,3dB} = 1200 \text{ rad/s}$. Using the short circuit time constant test, we start by opening the high frequency poles:

LF model



Applying the method, we can get the thevenin resistances and thus time constants for each capacitor. To simplify the calculations, we can ignore the effects of r_{o1} and r_{o2} here since they'll both be fairly large values:

$$R_{C_{C1}} = R_s + R_i + (R_{B2} // R_{B3} // r_{\pi1}) = 50 + 240 + (47000 // 56000 // 3738)$$

$$\rightarrow \tau_{C_{C1}} = C_{C1} \cdot 3550.6 \text{ s}$$

$$R_{C_{C2}} = R_C + R_L = 2400 + 50000$$

$$\rightarrow \tau_{C_{C2}} = C_{C2} \cdot 52400$$

For R_{C_E} we can ignore the effects of r_{o1} and r_{o2} to simplify the calculations since they'll be relatively larger and won't affect the results significantly. This greatly reduces the complexity of solving for the input resistance and we get:

$$R_{C_E} = R_E // \left[\frac{1}{\beta+1} (r_{\pi1} + R_{B2} // R_{B3} // (R_s + R_i)) \right] = 2400 // \left[\frac{1}{301} (3738 + 47k // 56k // 290) \right]$$

$$\rightarrow \tau_{C_E} = C_E \cdot 13.296$$

C_E is substantially smaller than the other capacitors, meaning it will dominate the low frequency pole, and so we can revolve our value around it. Our absolute limit is 1200 rad/s or about 191 Hz, so to be safe we solve for C_E and round up to the nearest common capacitor:

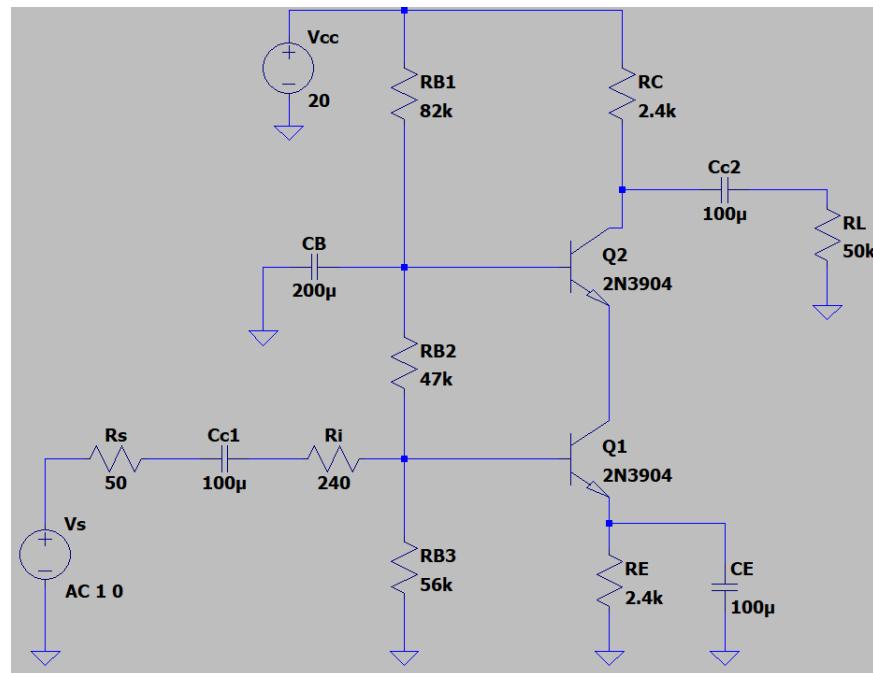
$$1200 \text{ rad/s} = \omega_{C_E} = \frac{1}{C_E \cdot 13.296}$$

$$\rightarrow C_E = 62.7 \text{ } \mu\text{F}$$

Rounding up to the nearest capacitor takes us up to $C_E = 100 \mu F$. We want to avoid something too large as from a design standpoint we want to use the smallest capacitor possible with a reasonable tolerance of at least 10%. This capacitance value fits that criteria, even if it does seem bigger than what we need

Since C_E dominates the pole, we can safely reuse this value for the other two capacitors, and so $C_E = C_{C1} = C_{C2} = 100 \mu F$.

This completes the LTSpice model, and we can begin the rest of the problem:



A. Measuring the DC point

We can easily find the dc point using the .op directive, resulting in:

```

--- Operating Point ---
V(n001):      20      voltage
V(n002):    15.0725   voltage
V(n003):    3.54203e-11 voltage
V(n004):    10.6703   voltage
V(n005):    9.99722   voltage
V(n006):      0      voltage
V(n007):    1.32361e-14 voltage
V(n008):    6.4857e-13 voltage
V(n009):    5.63239   voltage
V(n010):    4.95922   voltage
I(CB):      2.13406e-15 device_current
I(CE):      2.33083e-16 device_current
I(Cc1):     2.64722e-16 device_current
I(Cc2):     -7.08405e-16 device_current
I(RB1):     0.000113777 device_current
I(RB2):     0.000107189 device_current
I(RB3):     0.000100578 device_current
I(RC):      0.00205314 device_current
I(RE):      0.00206634 device_current
I(RL):      7.08405e-16 device_current
I(Ri):      2.64722e-16 device_current
I(Rs):      2.64722e-16 device_current
I(Vcc):     -0.00216692 device_current
I(Vs):      2.64722e-16 device_current
Ib(Q1):     6.61108e-06 device_current
Ib(Q2):     6.58747e-06 device_current
Ic(Q1):     0.00205973 device_current
Ic(Q2):     0.00205314 device_current
Ie(Q1):     -0.00206634 device_current
Ie(Q2):     -0.00205973 device_current
Is(Q1):      0      device_current
Is(Q2):      0      device_current

```

Value	V _{B2}	V _{B1}	V _{C2}	V _{E2,C1}	V _{E1}	I _{B1}	I _{B2}	I _{C1}	I _{C2}
Calculated	10.7	5.7	15	10	5	6.69μA	6.67μA	2.01mA	2.0mA
Simulated	10.67	5.63	15.07	9.997	4.96	6.61μA	6.59μA	2.06mA	2.05mA

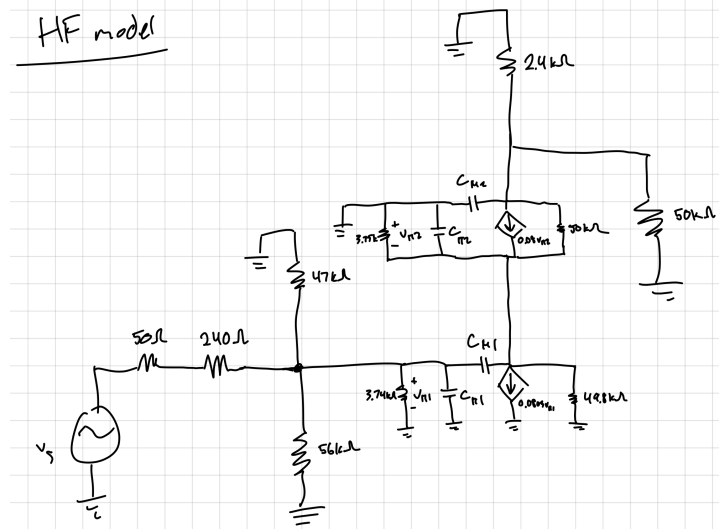
We can see that the values are very very close to one another. LTSpice gives more accurate results of course as its model is much more complex than the one we used, but we see that our simplifications provided for a very accurate result.

B. Bode plot frequency cutoff analysis

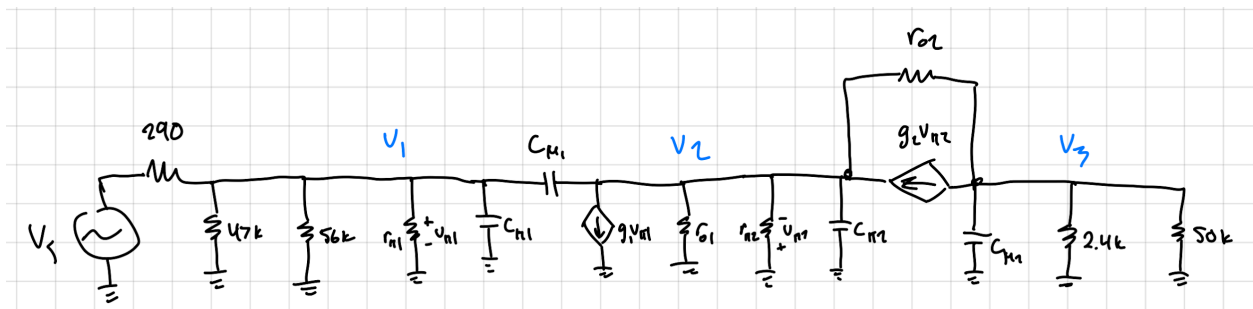
Earlier, we rounded up the capacitor to make sure we're well away from the 1200 rad/s limit for the pole, so we need to recalculate the pole with our new value to determine the pole. We only need to look at the pole from C_E since it will dominate. This gives

$$\omega_{L,3dB} = \omega_{C_E} = \frac{1}{100 \cdot 10^{-6} \cdot 13.296} = 120.7 \text{ Hz}$$

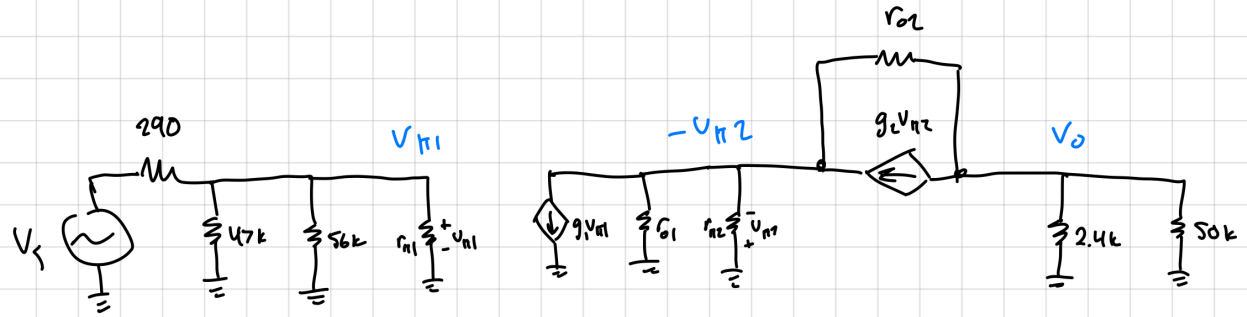
Let's quickly calculate the expected high frequency cutoff of the circuit before simulating. The high frequency circuit is:



We can rearrange the components to get:



Now we apply Miller's Theorem across $C_{\mu 1}$ to simplify the circuit and analyze the high frequency poles with our open circuit time constant method. Applying Miller's theorem we get the following circuit and equations:



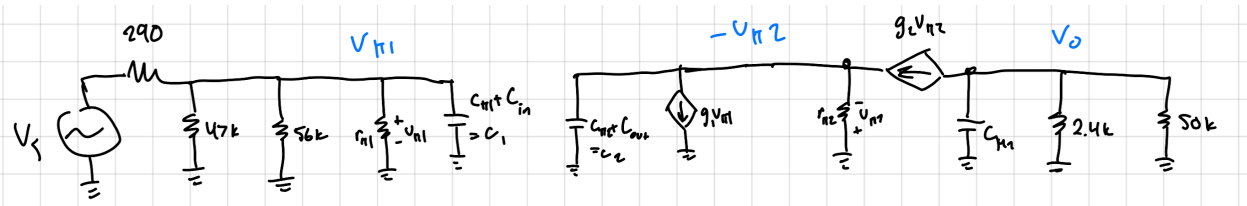
$$\frac{V_s - V_{\pi 1}}{290} = \frac{V_{\pi 1}}{47k // 56k // r_{\pi 1}}$$

$$\frac{0 - V_o}{2.4k // 50k} = \frac{V_o + V_{\pi 2}}{r_{o2}} + g_{m2} V_{\pi 2}$$

$$\frac{V_o + V_{\pi 2}}{r_{o2}} + g_{m2} V_{\pi 2} = g_{m1} V_{\pi 1} + \frac{-V_{\pi 2}}{r_{o1} // r_{\pi 2}}$$

Solving gives us a gain of $A_{C_{\mu 1}} = \frac{-v_{\pi 2}}{v_{\pi 1}} = -1.04511 \text{ V/V}$. This would give us $C_{\text{miller, in}} = C_{\mu 1}(1.045 + 1) = 8.1804 \text{ pF}$ and $C_{\text{miller, out}} = C_{\mu 1}(1.045 + 1)/1.045 = 7.8273 \text{ pF}$.

To simplify our calculations as much as possible, we can also ignore the effects of r_{o1} and r_{o2} for now since they're relatively larger and won't have a significant impact on the resulting calculation. With this our circuit becomes:



Now we apply the open circuit test. For $C_1 = C_{m, in} + C_{\pi 1} = 16.26 \text{ pF}$, the equivalent resistance is $R_{C1} = (R_s + R_i) // R_{B2} // R_{B3} // r_{\pi 1} = 266.3\Omega$, and we get $\omega_{C1} = \frac{1}{R_{C1} C_1} = 230.923 \text{ MHz}$

For $C_2 = C_{m, out} + C_{\pi 2} = 15.91 \text{ pF}$, we get

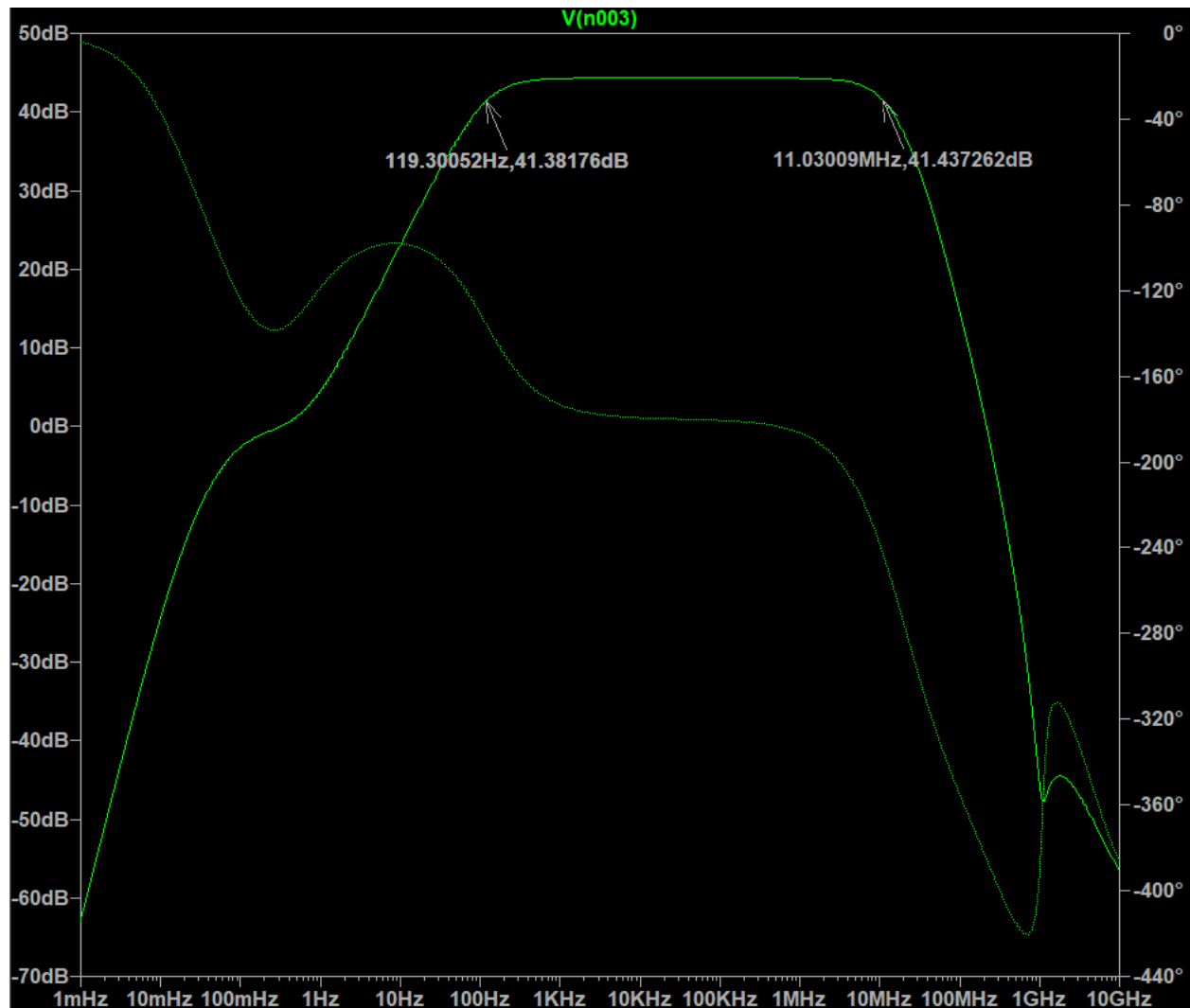
$R_{C2} = (R_s + R_i) // R_{B2} // R_{B3} // r_{\pi 1} = 266.3\Omega$, and we get $\omega_{C2} = \frac{1}{R_{C2} C_2} = 5.045 \text{ GHz}$

Lastly for $C_{\mu 2} = 4 \text{ pF}$, we get an equivalent resistance of $R_{c2} = 2400/50000 = 2.29 \text{ k}\Omega$ and the resulting pole is 109.15 MHz .

Both C_2 and $C_{\mu 2}$ dominate the pole fairly evenly,, so the pole will be split between them and we can find it like so:

$$\omega_{H,3dB} = \left(\frac{1}{\omega_{c2}^2} + \frac{1}{\omega_{\mu 2}^2} \right)^{\frac{-1}{2}} = \left(\frac{1}{109.15 \text{ MHz}^2} + \frac{1}{230.923^2} \right)^{\frac{-1}{2}} = 98.7 \text{ MHz}$$

Simulating the circuit results in the following Bode plot for v_o :



As is pointed out in the plot, the 3dB cutoff values give us that $\omega_{L,3dB} \simeq 119.3 \text{ Hz}$ and $\omega_{H,3dB} \simeq 11.0 \text{ MHz}$.

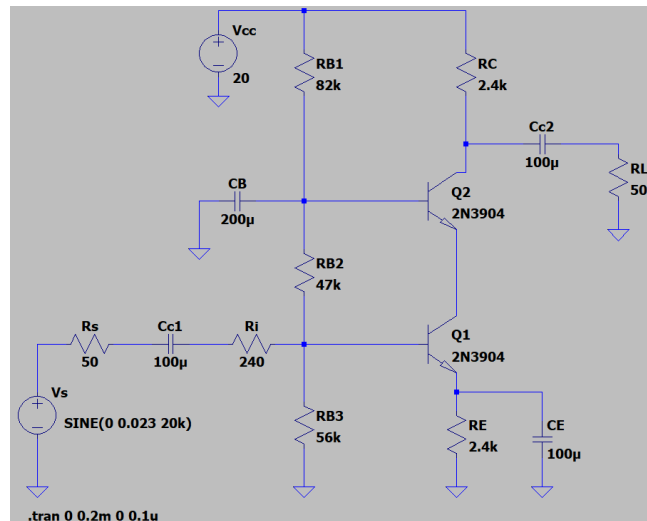
Comparing the calculated values with the simulated values:

Value	Calculated	Simulated
$\omega_{L,3dB}$	121 Hz	119 Hz
$\omega_{H,3dB}$	98.7 MHz	11.0 MHz

We observe that our low frequency cutoff is actually very accurate to what we had calculated. However, the high frequency cutoff is way

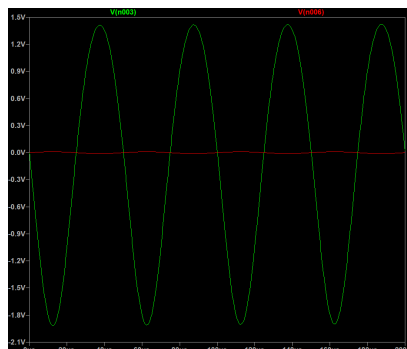
C. Midband Input Voltage Analysis

Observing the midband we can pick an arbitrary frequency for the analysis. I chose 20 kHz as it is right in the middle of the midband and is a nice round number. From here we can use the transient analysis feature.

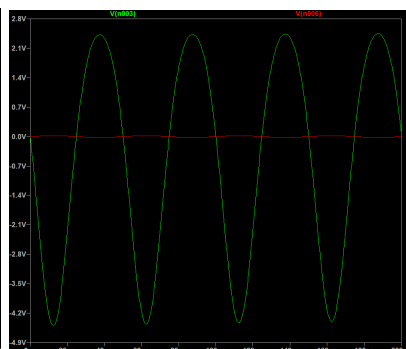


Adjusting the amplitude of the sinusoidal input gives the following:

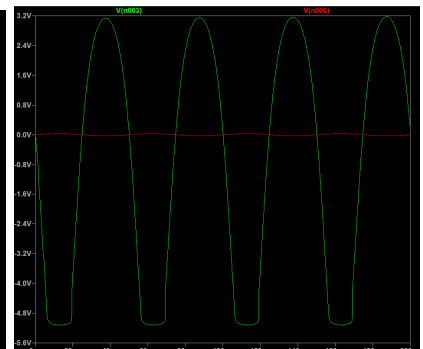
10mV



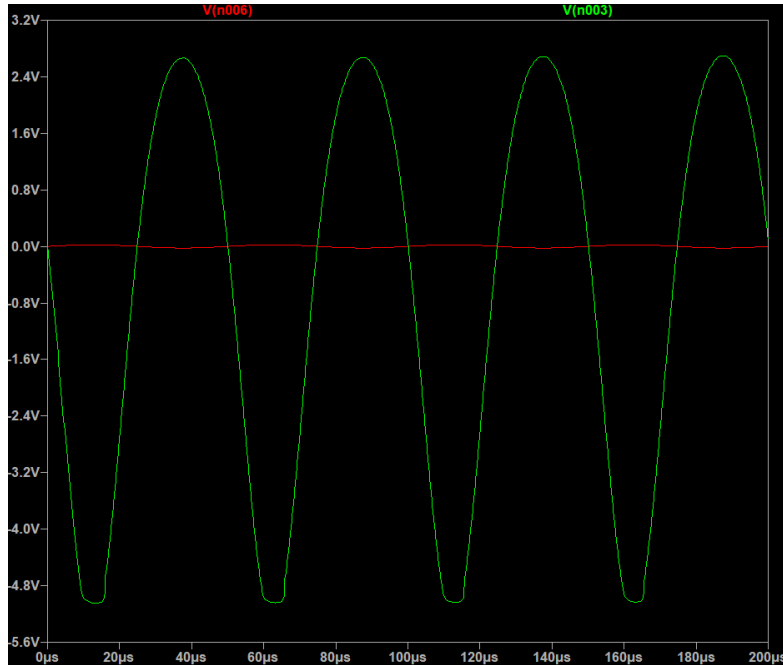
20 mV



30 mV



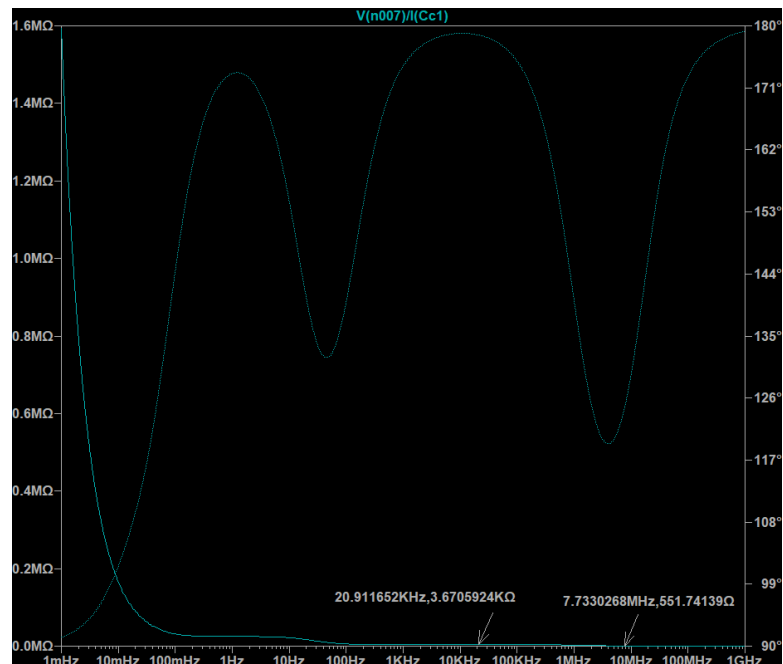
Adjusting the value to 30 mV distorts the signal and it is no longer linear. Further tuning shows that the signal begins to show more apparent signs of distortion between 22 and 23 mV:



23 mV distortion

D. Input Impedance Verification

To verify our design, we can plot the voltage vs the current at our input and see the result:



We see that at the midband, the input resistance bottoms out at roughly 3.6 k Ω , so just above the design requirement of 3.5 k Ω .

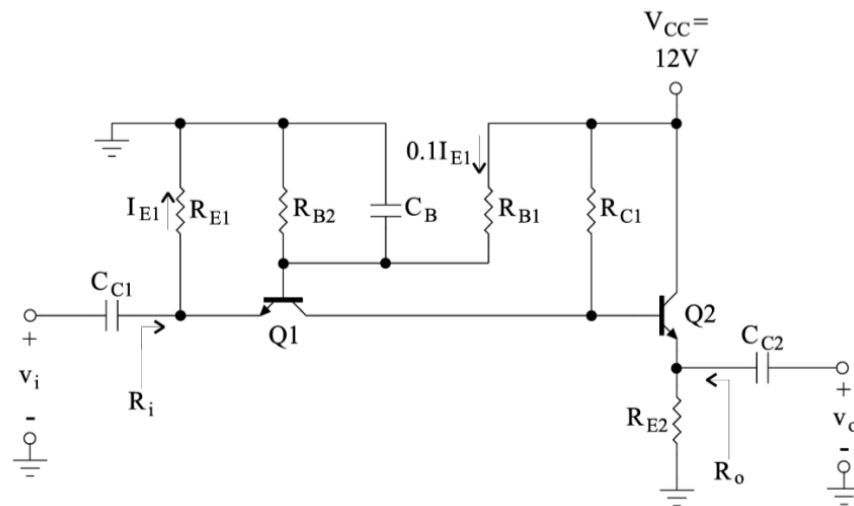
However, if we push it to the very edge of the midband, at around 7.7MHz, we see it drops significantly to 550 Ω . Clearly this is below the value we wanted, and it shows that the simplifications we use for analyzing the midband only works right in the middle of the midband, that our model starts to fall apart at the extremes. Once again, this proves that in the real world, it's much better to use a complex simulation for important design choices than to use simplified models and hand calculations.

Conclusion

We see that with the cascode amplifier our hand calculations provided a fairly reasonable model of the circuit, but it seemed to break down at our high frequency cutoff. Still, it makes for a rather accurate model with lower frequency calculations.

Problem 2

P2. The circuit shown below is a common-base followed by a common-collector amplifier stage. It is used as a repeater in an analog 50Ω coaxial cable system. It should be designed so that at midband it has an input impedance $R_i=50\Omega\pm5\Omega$, and a similar output impedance $R_o=50\Omega\pm5\Omega$. Choose 2N3904 for the transistors Q1 and Q2.

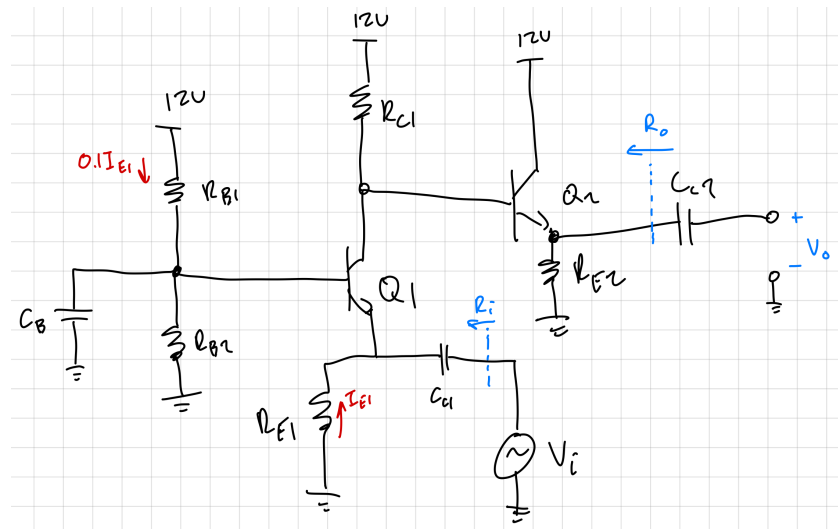


- Use $V_{CC}=12V$ and the $\frac{1}{3}$ rule such that $V_{E1}=V_{CC}/3$ and $I_1=0.1I_{E1}$ (as shown), the design criteria that at midband $R_i=R_o=50\Omega\pm5\Omega$, and that the low-frequency cut-in (the low-frequency 3dB point) occurs at, or below 1000Hz, to obtain design values for R_{E1} , R_{C1} , R_{B1} , R_{B2} , R_{E2} , C_{C1} , C_B and C_{C2} . (Hint: assuming that the source has an output impedance of 50Ω and that the repeater will be attached to a 50Ω load, C_{C1} and C_{C2} will both “see” similar resistance values - you can design them together to place the low frequency cut-in at 1000Hz and then design C_B so that it is as small as possible without changing the low frequency cut-in value).
- Using standard available resistor and capacitor values that are closest to your design values, implement the schematic and measure R_i and R_o at midband. Measure the midband gain $A_M=v_o/v_s$ of your amplifier, without a load resistance and without a source impedance (set $R_S=0$).
- Attach a voltage source with an output impedance of 50Ω to the amplifier input and a 50Ω load to the output. Obtain the amplitude and phase Bode plots. Find the low-frequency cut-in and the high-frequency cut-off points. Adjust any low frequency capacitors accordingly to meet the specifications.

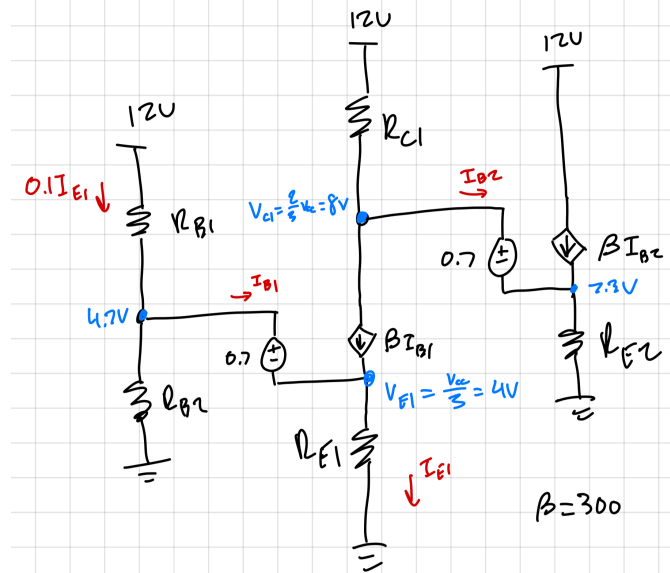
In your report, discuss your observations, compare your simulation results with your computation, and discuss any gained insights.

A. Design Values

Let's first begin by redrawing the circuit into a more digestible format:



Now we get the DC model using our DC model of the BJT. Like in question 1 the value of β is 300 for the 2N3904 transistor. We can also go ahead and apply the $\frac{1}{3}$ rule on Q1:



We develop the following equations:

$$\frac{12-4.7}{R_{B1}} = 0.1I_{E1} = 0.1 \frac{4}{R_{E1}} = \frac{0.4}{R_{E1}} \rightarrow \frac{7.3}{R_{B1}} = \frac{0.4}{R_{E1}}$$

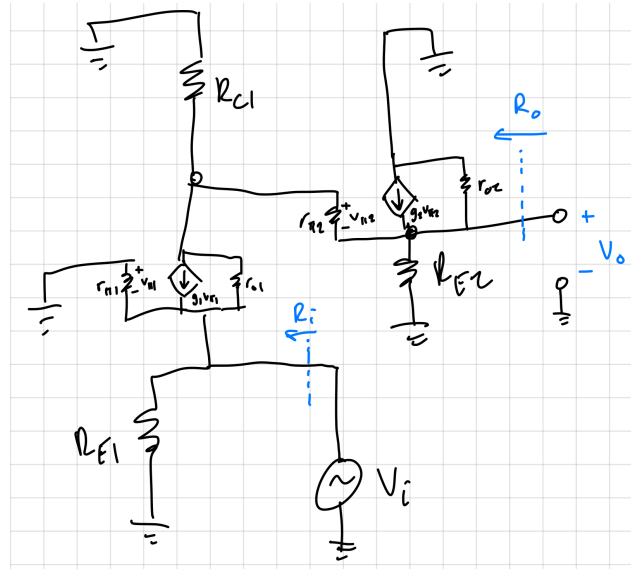
$$\frac{12-4.7}{R_{B1}} = I_{B1} + \frac{4.7}{R_{B2}}$$

$$(300 + 1)I_{B1} = \frac{4}{R_{E1}} \rightarrow I_{B1} = \frac{4}{301 \cdot R_{E1}}$$

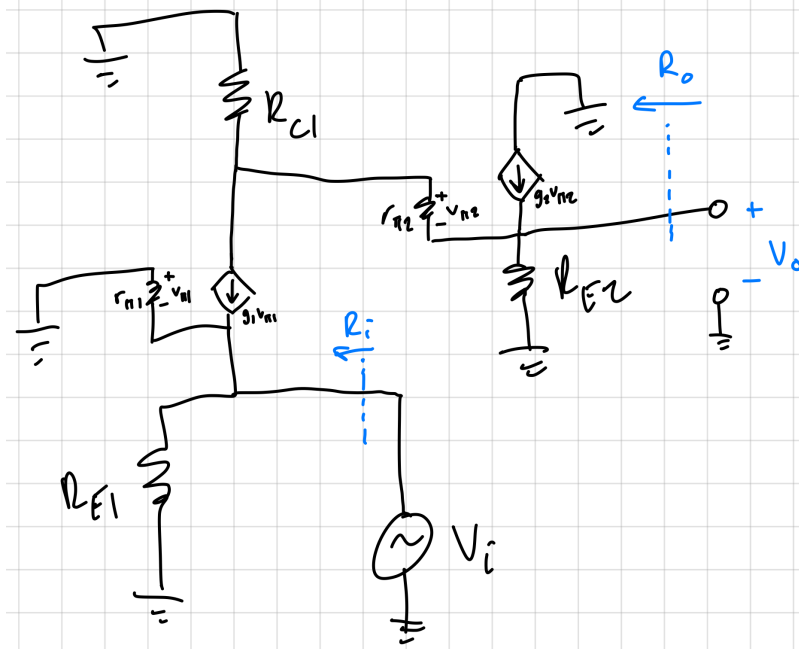
$$\frac{12-8}{R_{C1}} = I_{B2} + 300I_{B1}$$

$$(300 + 1)I_{B2} = \frac{8-0.7}{R_{E2}} \rightarrow I_{B2} = \frac{7.3}{301 \cdot R_{E2}}$$

We have 5 equations and 7 unknowns. To get the rest of the expressions, let's take a look at the midband circuit and figure out what it would take to meet our input and output resistance requirements. We'll substitute the transistors for our small signal model as we do this:



Since we know r_{o1} and r_{o2} are both relatively large values, they won't have a significant effect on the circuit and we can simplify our calculations by neglecting them. This gives us the following:



Where $r_{\pi1} = \frac{V_T}{I_{B1}}$, $r_{\pi2} = \frac{V_T}{I_{B2}}$, $g_1 = \frac{\beta I_{B1}}{V_T}$, and $g_2 = \frac{\beta I_{B2}}{V_T}$. We assume $V_T = 25\text{mV}$ @ 25°C .

With this circuit we can use our trick for the controlled current sources and get the equivalent resistances to be:

$$R_i = R_{E1} // \left(\frac{1}{\beta+1} \right) r_{\pi1}$$

$$R_o = R_{E2} // \left(\frac{1}{\beta+1} \right) (r_{\pi2} + R_{C1})$$

From our design specifications, we want both of these resistances to be roughly equal to 50 Ω . While not immediately intuitive, we can actually play around with R_i , and in particular, R_{E1} to figure out another expression. Looking at the current through R_{E1} we observe that

$\frac{4}{R_E} = I_{E1} = (\beta + 1)I_{B1}$. Rearranging this with $\beta = 300$ yields:

$$R_E = \frac{4}{301 \cdot I_{B1}}$$

Recall that $r_{\pi1} = \frac{0.025}{I_{B1}}$, and that we have a common denominator among these values.

Therefore, if we set $R_i = 50$ and put in these expressions, we end up with an equation for I_{B1} :

$$R_i = 50 = R_{E1} // r_{\pi1} \left(\frac{1}{\beta+1} \right) = \frac{4}{301 \cdot I_{B1}} // \left(\frac{0.025}{I_{B1}} \right) \left(\frac{1}{301} \right)$$

Solving for I_{B1} then gives us:

$$I_{B1} = 1.651 \mu A$$

With that we can also solve for $r_{\pi1}$, R_{E1} , I_{E1} , and the rest of the Q1 small signal parameters.

$$r_{\pi1} = 15.144 k\Omega$$

$$I_{E1} = 301 I_{B1} = 496.89 \mu A$$

$$R_{E1} = 8.050 k\Omega$$

For I_{B2} and R_{E2} , we can do something similar with the output resistance given the expression

$I_{B2} = \frac{7.3}{301 \cdot R_{E2}}$ that we got from our DC model to get:

$$R_o = R_{E2} // \left(\frac{1}{\beta+1} \right) (r_{\pi2} + R_{C1})$$

$$\rightarrow 50 = R_{E2} // \left(\frac{1}{301} \right) \left(\frac{0.025}{\left(\frac{7.3}{301 \cdot R_{E2}} \right)} + R_{C1} \right)$$

We can combine this expression with the equations we found in the DC circuit to calculate the remaining values. We calculate the required resistor values to be:

$$R_{B1} = 146.912 k\Omega$$

$$R_{B2} = 97.838 k\Omega$$

$$R_{C1} = 8.020 k\Omega$$

$$R_{E1} = 8.050 k\Omega$$

$$R_{E2} = 6.926 k\Omega$$

And the resulting base currents used for finding small signal parameters:

$$I_{B1} = 1.651 \mu A$$

$$I_{B2} = 3.502 \mu A$$

$$I_{E1} = 496.89 \mu A$$

From that we also get our small signal parameters:

$$r_{\pi1} = \frac{V_T}{I_{B1}} = 15.144 k\Omega$$

$$r_{\pi 2} = \frac{V_T}{I_{B2}} = 7.139 \text{ k}\Omega$$

$$g_1 = \frac{\beta I_{B1}}{V_T} = 0.01981$$

$$g_2 = \frac{\beta I_{B2}}{V_T} = 0.04202$$

We need to convert the resistances to standard values as per the requirement, so we end up with:

$$R_{B1} = 146.912 \text{ k}\Omega$$

$$R_{B2} = 97.838 \text{ k}\Omega$$

$$R_{C1} = 8.020 \text{ k}\Omega$$

$$R_{E1} = 8.050 \text{ k}\Omega$$

$$R_{E2} = 6.926 \text{ k}\Omega$$

R_{B1}	150 k Ω
R_{B2}	100 k Ω
R_{C1}	8.2 k Ω
R_{E1}	8.2 k Ω
R_{E2}	6.8 k Ω

As a sanity check, let's recalculate the values of the input and output resistances to ensure we're still within the range:

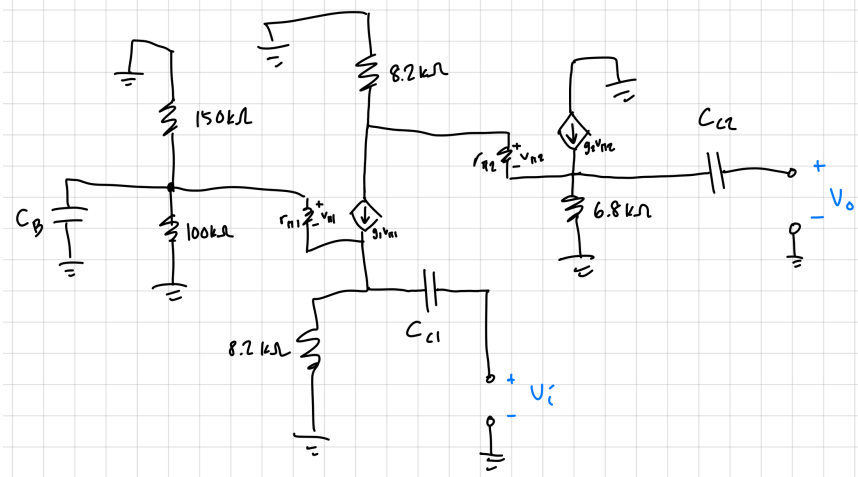
$$R_i = R_{E1} // \left(\frac{1}{\beta+1}\right)r_{\pi 1} = 50.006 \Omega$$

$$R_o = R_{E2} // \left(\frac{1}{\beta+1}\right)(r_{\pi 2} + R_{C1}) = 45.997 \Omega$$

We see that we're still within 5 Ω of 50 Ω for both values, so we're in the clear.

Finding the capacitor values with the low frequency cutoff

First we draw the low frequency model of the circuit with our newfound resistor values:



As described in the problem, C_{C1} and C_{C2} both effectively “see” the same circuit, so we’ll count them as one pole. The resistances they see are equivalent to R_i and R_o , which we had recalculated with the new values earlier, giving us $R_{C_{C1}} = 50.006 \Omega$ and $R_{C_{C2}} = 45.997 \Omega$. We can solve for the pole of the smaller value (C_{C2}) to be safe by finding the capacitance needed to set the cutoff at 1000 Hz or $2000\pi \text{ rad/s}$:

$$\omega_{C_{C2}} = \frac{1}{R_{C_{C2}} C_{C2}} = 2000\pi \quad \rightarrow C_{C2} = 3.46 \mu F$$

The two nearest standard values are $3.3 \mu F$ and $4.7 \mu F$. Since 1000 Hz is our maximum cut-in frequency, we will have to round up as going lower can set us over that limit. Thus we get:

$$C_{C1} = C_{C2} = 4.7 \mu F$$

Which results in a cut-in frequency of $\omega_{C_{C2}} = \frac{1}{R_{C_{C2}} C_{C2}} = 4625.7 \text{ rad/s} = 736.2 \text{ Hz}$

Now for C_B , we find the equivalent resistance of the circuit to be:

$$R_{C_B} = 100k // 150k // (\beta + 1)r_{\pi 1} = 59220.5 \Omega$$

This value is much larger than that seen by C_{C2} and C_{C1} , we can play around with this value to one that doesn’t significantly affect the resulting cutoff frequency.

Using $\omega_{L,3dB} = \sqrt{\omega_{C_{C1,C2}}^2 + \left(\frac{1}{R_{C_B} C_B}\right)^2} = 4625.7$ we get $C_B = 1.05 \mu F$. Rounding up gets us:

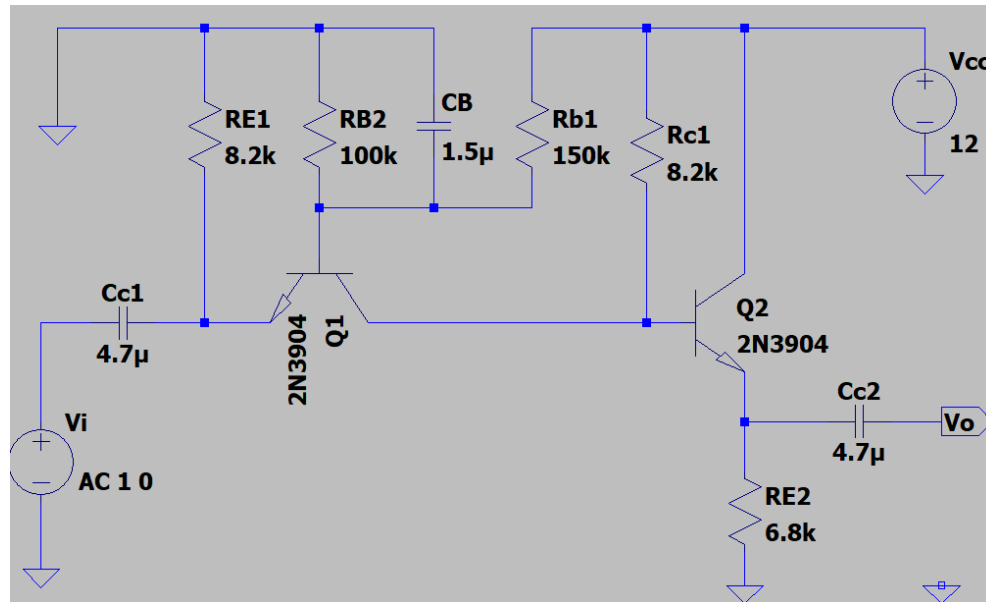
$$C_B = 1.5 \mu F$$

Thus for our capacitors we have:

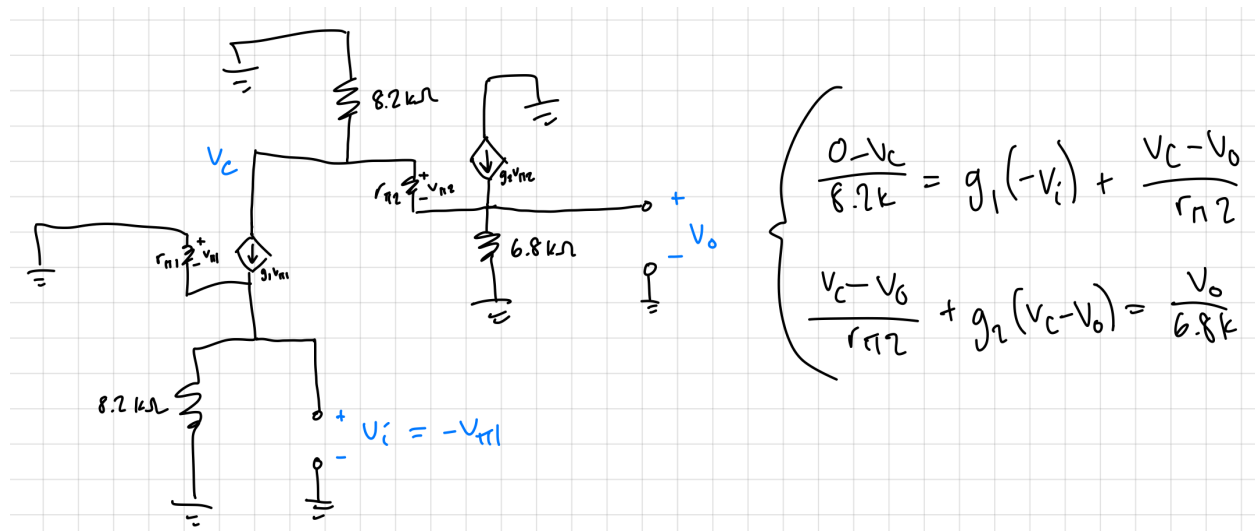
C_{C1}	$4.7 \mu F$
C_{C2}	$4.7 \mu F$
C_B	$1.5 \mu F$

B. Midband Analysis of R_i , R_o , and A_M

Implementing the values we got in part A gives us the following circuit in LTspice:



For our comparisons later, we should first determine the midband gain. The midband circuit with our values is:



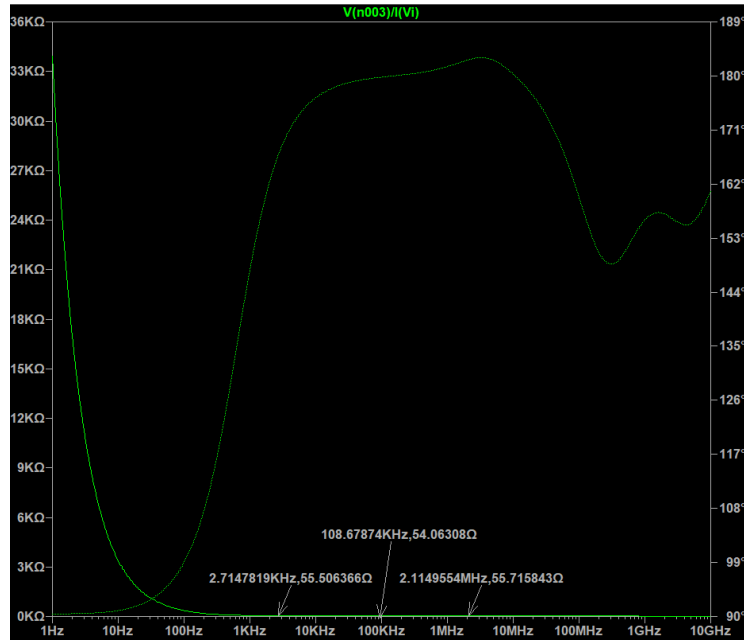
Midband circuit with equations

The above diagram already includes the two equations we need to solve. Using CAS and

solving for $A_M = \frac{v_o}{v_i}$ we get that $A_M = 161.23 \text{ V/V}$

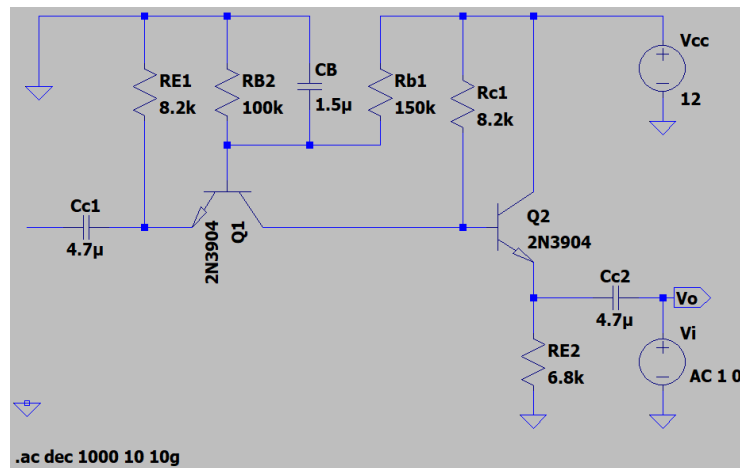
With that we can use LTspice to plot the simulated values and compare the results.

For R_i :

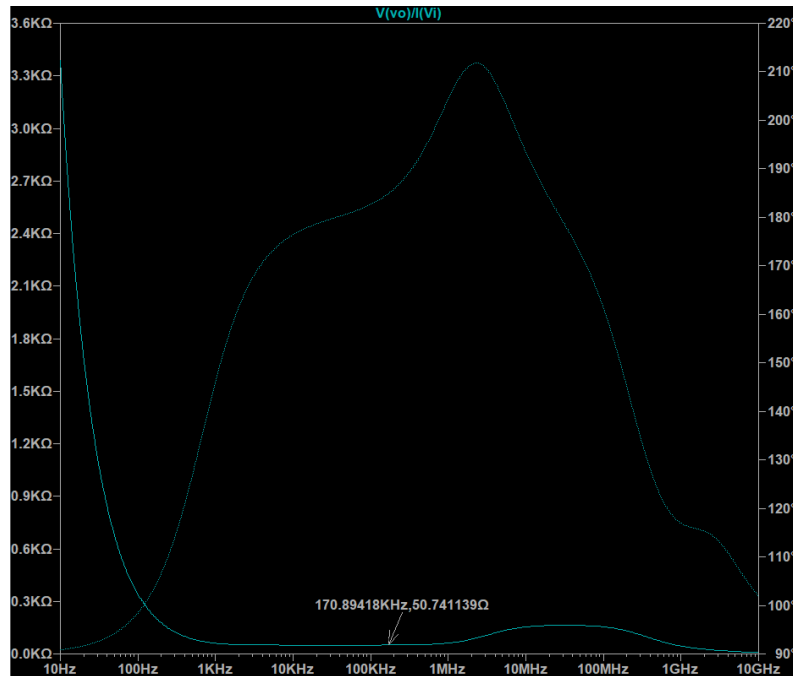


At a midband frequency of around 100 kHz, we find that $R_i = 54.1 \Omega$ which is within our maximum of 55Ω. However, when we approach either cutoff, we see that this input resistance rises above the 55Ω limit we had set.

For R_o we move the voltage source to the output like so:

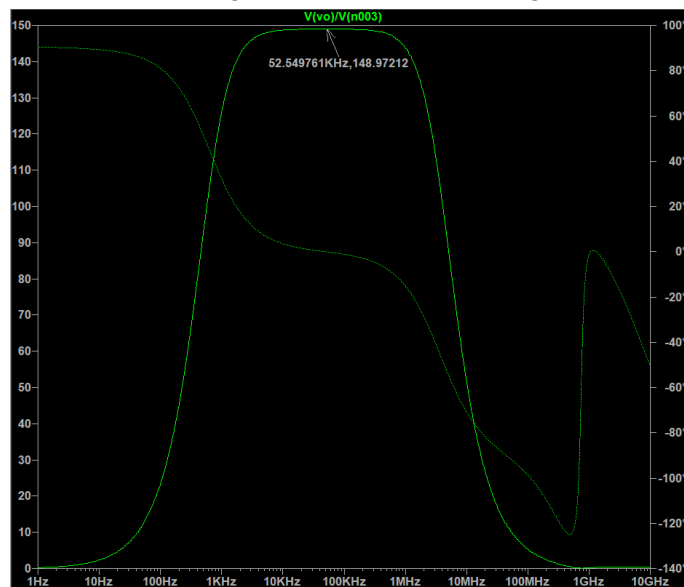


From this we get the following plot:



We get $R_o = 50.7 \Omega$ at the midband, around 170 kHz.

For the midband gain we move the voltage source back to the input and get:



We see that at around 52kHz, we get a midband gain of 148.97.

A table comparing the values we calculated and the ones we found in the simulation:

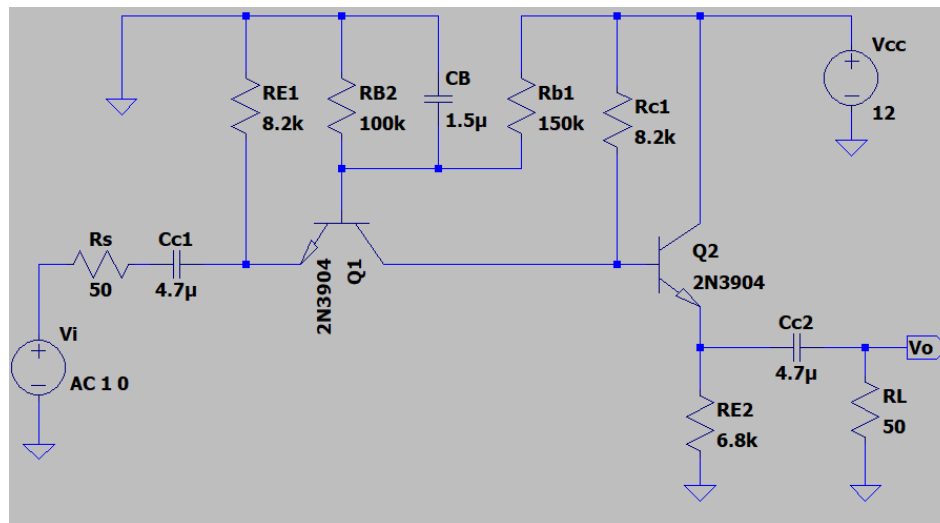
Value	Calculated	Simulated	Error
R_i	50.006 Ω	54.1 Ω	7.58%

R_o	45.997 Ω	50.7	9.27%
A_M	161.23	148.97	7.60%

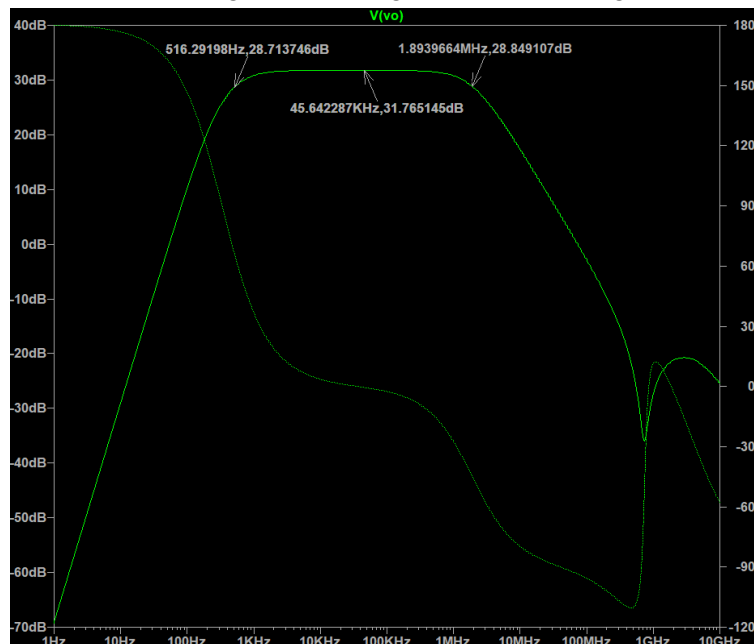
We see that our calculations have a fairly consistent margin of error of a little under 10%. Given all of our simplifications throughout the calculations, this is quite reasonable and not too bad.

C. Adjusting Capacitors for Input Resistance & Output Load

Updating the LTspice circuit gives us the following:

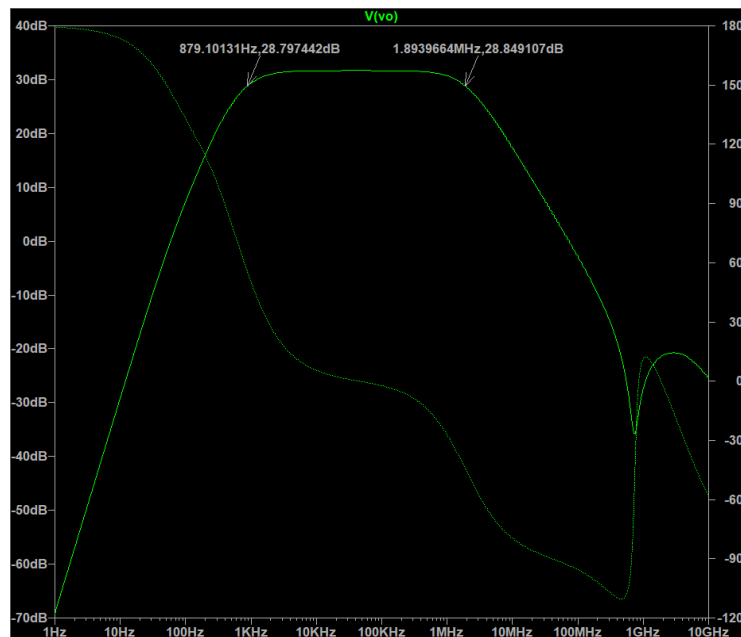


Running the simulation and checking the output gives the following Bode plot:



We get the cutoff points $\omega_{L,3dB} = 516.3 \text{ Hz}$ and $\omega_{H,3dB} = 1.894 \text{ MHz}$.

We want to raise the frequency so that we get as close to 1000 Hz as possible. Since C_{C1} and C_{C2} are dominant and have a small thevenin resistance, any changes will lead to drastic results and make fine-tuning difficult, so we'll instead focus on C_B and sequentially adjust it to get somewhat close to the 1 kHz. Through some trial and error we finally get $R_B = 0.015 \mu F$ with the Bode plot shown below:



The next smaller standard value would be $R_B = 0.01 \mu F$, but through testing we see that this would actually overshoot the cutoff frequency by about 30 Hz, and so this is as close as we can get with the standard capacitor values. Thus our adjusted capacitors are:

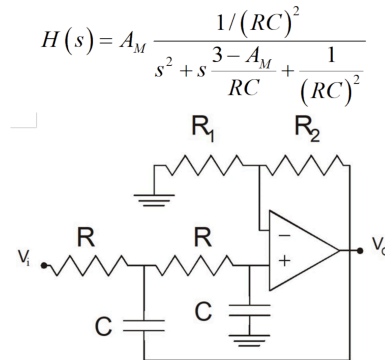
C_{C1}	$4.7 \mu F$
C_{C2}	$4.7 \mu F$
C_B	$0.015 \mu F$

Conclusion

We observe that our simplified model for hand calculations provides us with enough information to begin designing a proper circuit and provides reasonable estimation for the value of the components that should be used. We successfully created the circuit based on the requirements, however we note that in the midband as the frequency approaches the cutoffs, we see R_i and R_o exceed the set limits and should keep that in mind and make adjustments if we were to actually use this exact circuit for the engineering task we were given.

Problem 3

P3. The following circuit is used as a 2nd order low-pass active filter. The transfer function is:



Here, A_M is the pass-band gain (and also the DC gain), given by $A_M = 1 + R_2/R_1$. The transfer function has two poles. By varying A_M , you can also control the damping ratio, so that the two poles can be made complex. This low-pass filter can be made into a 2nd order Butterworth filter by placing the poles at specific locations on a circle, centered at the origin in the s -plane.

Using an UA741 as the op-amp and a $\pm 15V$ power supply, introduce the circuit in your Spice program, setting R and $R_1 + R_2$ equal to $10k\Omega$

- Calculate the values of C and A_M that will make the filter a 2nd order low-pass Butterworth filter with a 3dB cut-off frequency of 10kHz. Plot the magnitude and phase Bode plots of the circuit for the computed values for C and A_M , and show where your poles are located in the s -plane.
- Keeping $R_1 + R_2 = 10k\Omega$, change R_1 and R_2 slowly to increase A_M . At what value of A_M does your circuit start to oscillate? (You can check for oscillations by removing the source and grounding the input). Plot the output v_o when the circuit starts to oscillate and report the R_1 , R_2 values at this point. At what value of A_M did you expect your circuit to start to oscillate? Why? Measure and report the oscillation frequency. Describe what is happening to your poles in the s -plane when you change A_M (i.e. draw the root locus for your active filter and explain the behavior).

A. Calculating C & A_M for Butterworth filter

For a second order circuit such as this one, the transfer function takes the shape of:

$$H(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

For this particular circuit with our known values, we get a cut-off frequency of $\omega_c = \frac{1}{RC} = \frac{1}{C \cdot 10^4}$ and a damping ratio of $\xi = \frac{3-A_M}{2}$.

To find C, we're already given the 3dB cut-off frequency of 10 kHz, so we have a simple equation to solve:

$$\omega_c = 10000 \cdot 2\pi = \frac{1}{10000 \cdot C}$$

This gives us:

$$C = 1.5915 \text{ nF}$$

To get the gain, we look to the table of normalized Butterworth polynomials and get that, for a second order Butterworth filter, the normalized polynomial is $s^2 + s\sqrt{2} + 1$ (refer to *Lecture Slideset 29 pg. 18*). This tells us that $2\xi = \sqrt{2}$, and plugging in the expression for the damping ratio from earlier gives:

$$2\left(\frac{3-A_M}{2}\right) = \sqrt{2}$$

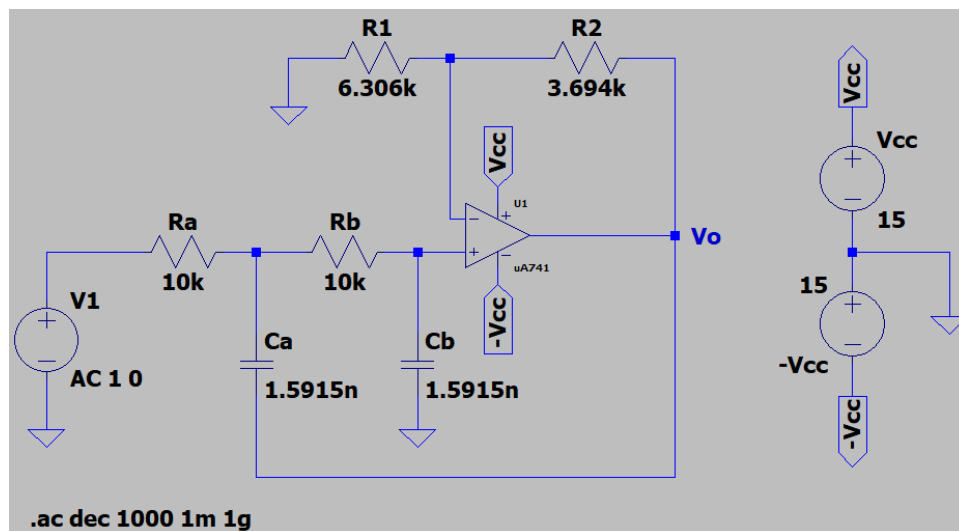
Solving for A_M we find that $A_M = 3 - \sqrt{2} = 1.586$

Given that $A_M = 3 - \sqrt{2} = 1 + \frac{R_2}{R_1}$, as well as that $R_1 + R_2 = 10 \text{ k}\Omega$, we can solve for R_1 and R_2 to get:

$$R_1 = 6.3060 \text{ k}\Omega$$

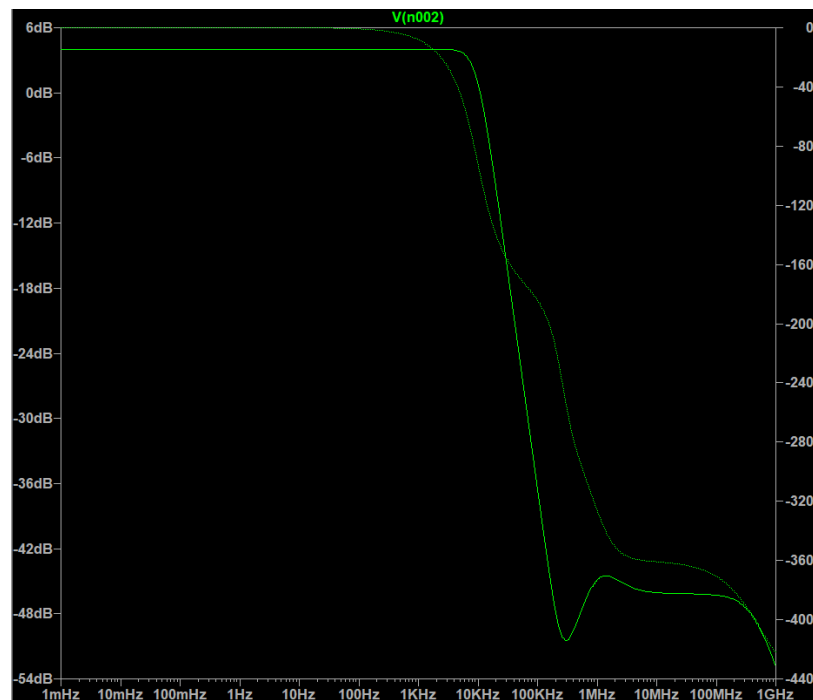
$$R_2 = 3.6940 \text{ k}\Omega$$

Putting all this into LTspice:



We can get our UA741 model from the link provided in the top comment of this forum: <https://forum.allaboutcircuits.com/threads/how-to-add-op-amp-741-to-ltspice.192843/> There is a minor error on line 17 that we have to fix which requires Altium, essentially we remove the

unnecessary commas and get “*egnd 99 0 poly(2)(3,0)(4,0) 0 .5 .5*” With this fix we can simulate the circuit. Plotting the output voltage gives the following Bode plot for both magnitude and phase:



We already know that for a Butterworth filter, the poles are 45° left of the imaginary axis, but we can easily calculate the actual poles by solving for the roots of the denominator

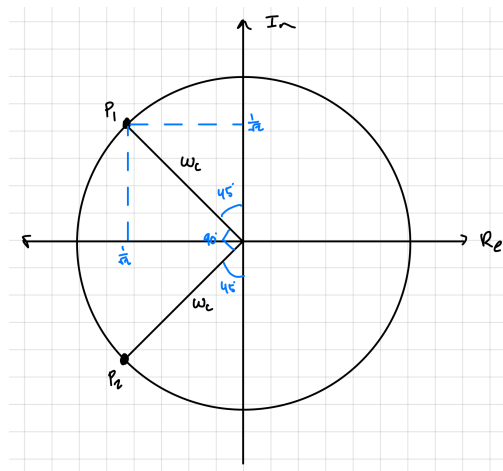
$$s^2 + s \frac{3-A_M}{RC} + \left(\frac{1}{RC}\right)^2 \text{ or } s^2 + s(3 - (3 - \sqrt{2}))(20000\pi) + (20000\pi)^2 \text{ to get:}$$

$$p_1 = -44429 + j44429, \text{ and } p_2 = -44429 - j44429$$

or rather

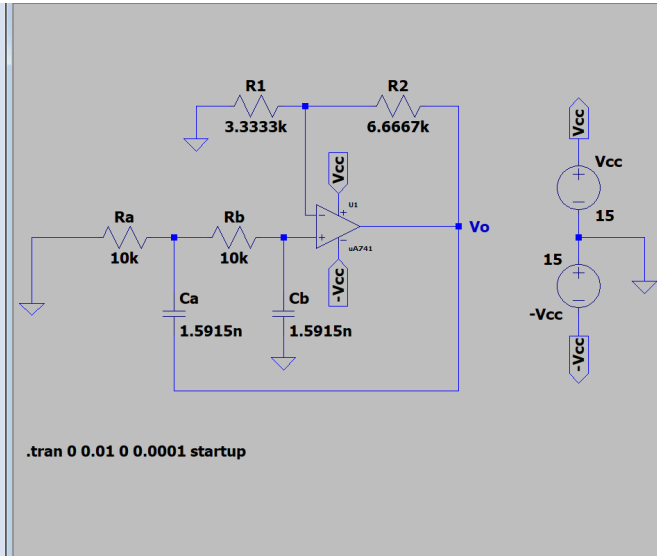
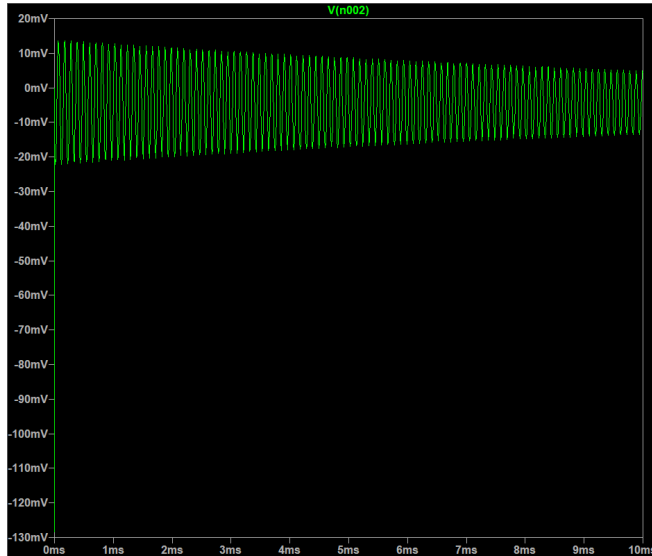
$$p_1 = \frac{\omega_c}{\sqrt{2}}(-1 + j), \text{ and } p_2 = \frac{\omega_c}{\sqrt{2}}(-1 - j)$$

Plotted out this looks like:



B. R_1 & R_2 Adjustment for A_M Oscillation

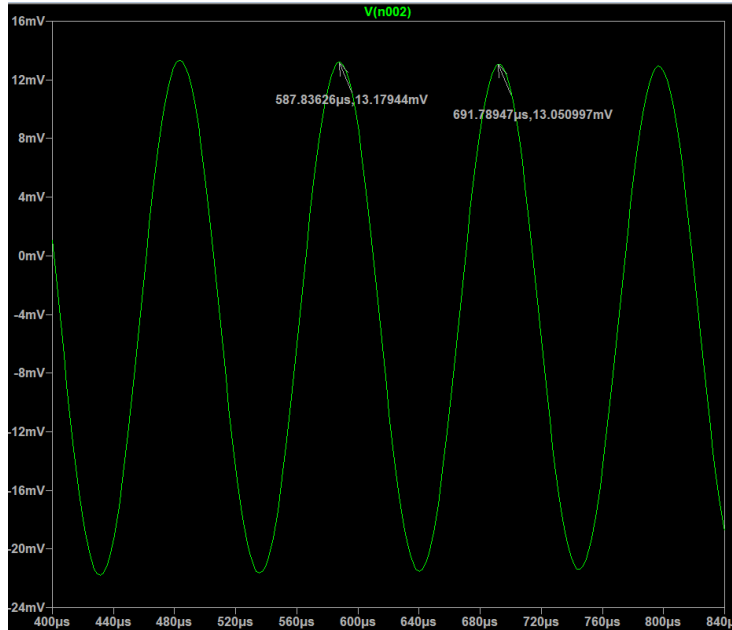
Adjusting R_1 and R_2 , the circuit begins to oscillate when $R_1 = R_1 = \frac{10000}{3} \Omega$ and $R_2 = \frac{20000}{3} \Omega$.



This is exactly what we should expect. If we adjust A_M , the circuit will begin to oscillate when the angle between our poles and the imaginary axis reaches zero, as in when $p_1 = j\omega_c$ and $p_2 = -j\omega_c$. This puts the system into marginal stability which creates the oscillation. This corresponds to when the damping ratio is set to zero ($\xi = 0$), as that means nothing is preventing oscillation and the circuit will oscillate indefinitely once disturbed.

Solving for $\xi = \frac{3-A_M}{2} = 0$ tells us that at $A_M = 3$, the circuit will begin to oscillate. Past this point the damping ratio becomes negative, and so the circuit will amplify the signal uncontrollably, until it reaches the op-amp limit which is set by the power supply, so ± 15 V.

In LTspice we can easily zoom in and get the period to find the frequency:

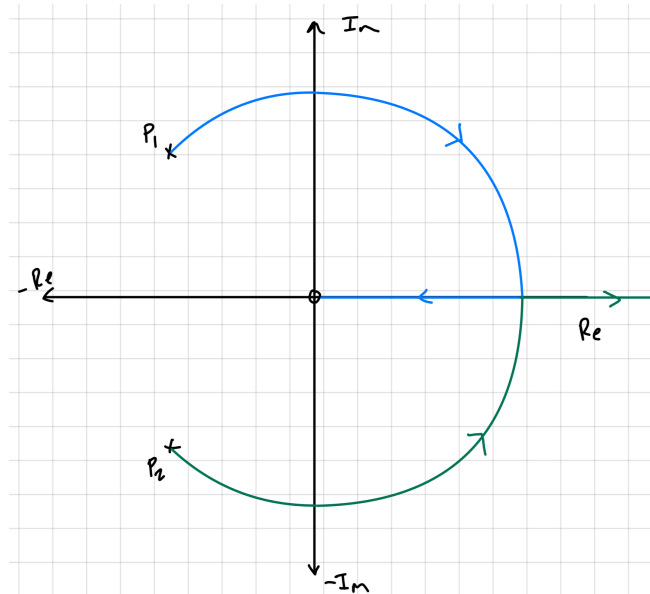


We get $\Delta\tau = 691.78947 - 587.83626 \mu s$ which corresponds to a frequency of

$$f = 9619.7 \text{ Hz}$$

We notice that this is very very close to the cut-off frequency we set of $\omega_c = 10 \text{ kHz}$, which is the value we expect it to oscillate at, and it's likely only off because we didn't put more digits to get the perfect ratio of 1:2 between R_1 and R_2 .

When we change A_M , the poles will move along root locus as shown below:



p_1 will move along the blue line and p_2 will move along the green line, and both will be horizontally aligned as they do so until they reach the horizontal axis, where they will split away from each other along the horizontal axis as one pole approaches zero while the other approaches infinity.